

6502 Monitor - ESP32 Pin Mapping (Atari 130XE Validated)

Atari ESP32 Project

June 12, 2026

1 Overview

Final pin assignment for ESP32 NodeMCU DevKit V1. Includes Atari RESET control to ensure synchronized startup.

2 Data Bus (Bank 0)

6502 Signal	ESP32 GPIO	Board Label
D0-D7	GPIO 4, 5, 13, 14, 16, 17, 18, 19	D4, D5, D13, D14, RX2, TX2, D18, D19

3 Address Bus

6502 Signal	ESP32 GPIO	Board Label
A0-A3	GPIO 34, 35, 36, 39	D34, D35, VP, VN (Input Only)
A4-A7	GPIO 32, 33, 21, 27	D32, D33, D21, D27

4 Control Signals

6502 Signal	ESP32 GPIO	Board Label	Description
PHI2	GPIO 2	D2	System Clock (IN)
R/W	GPIO 15	D15	Read/Write (IN)
SEL_N	GPIO 22	D22	D1XX_N / CCTL_N (IN)
ROMSEL	GPIO 23	D23	\$D800-\$DFFF Range (IN)
RAMSEL	GPIO 26	D26	\$D600-\$D7FF Range (IN)
EXTSEL	GPIO 3	RX0	Disable Atari Memory (OUT)
VCS	GPIO 25	D25	Device Select (OUT)
MPD	GPIO 0	BOOT	Math Pack Disable (OUT)
RESET	GPIO 12	D12	Atari RESET Control (OUT)

5 Hardware Safety & Boot Sequence

- **Atari RESET (GPIO 12):** Held LOW immediately on power-up to keep Atari paused.
- **ESP32 Boot (GPIO 0):** Safe boot via Atari's internal pull-up.